

SHUVAM PAL

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Education

Dayananda Sagar College of Engineering

Bengaluru, India

B.E. Artificial Intelligence and Machine Learning

Experience

Bentham AI 

Dec 2025 – Feb 2026

Backend Engineering Intern: Node.js, JavaScript, Go, Docker, GCP, GitHub Actions

Remote

- Built an AI-assisted workflow for structured company-registration drafts, using public-site API lookups, operator validation, retries, and automation handoff across multi-step filings.
- Reduced manual filing time by **85%** by implementing session recovery, browser automation, state validation, and failure-aware backend execution.
- Containerized services with Docker, deployed to Google Cloud Run, and automated releases through GitHub Actions, improving deployment velocity by **70%**.

Google Summer of Code 

May 2025 – Sept 2025

Mentee at Invesalious: Python, VTK, wxPython, NumPy, PyTorch, ONNX

Remote

- Built scientific Python tooling for MRI segmentation across **95 anatomical regions**, handling preprocessing, orientation correction, deep-learning inference, and generated-mask workflows.
- Reduced runtime by **87%** using asynchronous and parallel execution over large medical-image volumes while preserving inspectable outputs.
- Improved 3D reconstructed surface visualization and debugged data/model-output issues inside a mature open-source medical-imaging codebase.

Open Source

[Contributions](#) 

- Contributed to research-adjacent open source including **Invesalious** medical-imaging workflows, **IOOS** scientific-data tooling, and **Tiled** map/visual editor code, plus compatibility and security-validation fixes.

Projects

Hyoka | *Python, FastAPI, PostgreSQL, OpenTelemetry, Typer, JSON Schema, Docker*

- Built autonomous self-improvement cycles that mine failed evaluation slices, generate candidate patch bundles, launch validation runs, compare baseline versus candidate behavior, and record gated promotions.
- Designed reproducible replay and evaluation around immutable inputs, content-addressed artifacts, signed run manifests, audit history, JSON schema checks, latency/cost constraints, and LLM-judge hooks.

Penny Lane Capital | *Python, Agno, Pydantic, SQLite, Vector Memory, yfinance, JSONL*

[GitHub](#) 

- Built a research workflow with role-specific agents, debate, risk review, portfolio approval, checkpointed state, and replayable JSONL traces for repeated market windows.
- Evaluated decisions with historical-market backtests, reward-loop style scorecards, validation harnesses, and baseline comparisons across return, drawdown, and win-rate signals.

Seaweed: Contest Platform | *Go, React, PostgreSQL, Judge0, Redis, AWS, Docker, Kubernetes*

[GitHub](#) 

- Architected a full-stack evaluation platform supporting **500+ concurrent users**, automated judging, live rankings, and reproducible submission/result records.
- Designed async Go dispatch around Judge0 with PostgreSQL result schemas, Redis-backed runtime state, Docker/Kubernetes deployment, and **sub-5s P95 verdict latency**.

Technical Skills

Languages: Python, C++, Go, TypeScript, SQL

Frameworks: PyTorch, FastAPI, Node.js, React, VTK, wxPython, Agno

Tools: NumPy, PostgreSQL, SQLite, Redis, Vector Memory, OpenTelemetry, Docker, GitHub Actions, ONNX, JSONL